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Absence of female preference and the origin of a unisexual species, the Amazon molly (*Poecilia formosa*)

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Abstract

The role of hybridization as a formative process in evolution has received much attention in the past few decades. A particularly fascinating outcome of hybrid speciation is the formation of asexual hybrid species. This is the case in the Amazon molly (*Poecilia formosa*). It is of hybrid origin and had a *P. mexicana* female as mother and a *P. latipinna* male as father. This implies that at the beginning of the existence of Amazon mollies, a heterospecific mating must have occurred, indicating a breakdown of any potential prezygotic isolation between parental species. We studied the female mate choice behavior with present day *P. mexicana* and *P. latipinna* from several populations using standard binary choice tests. Overall, our results show that in extant *P. mexicana* and *P. latipinna* no female choice for conspecific males was detectable. Consequently, heterospecific matings are possible, even likely, and not hindered by any apparent behavioral prezygotic isolation. If female preferences in these species were comparable around the time the Amazon molly originated as a hybrid species ca. 100.000 years ago, matings leading to hybrids would be very likely. Furthermore, for the first time we tested the mating preferences of F₁ hybrid

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females and found that they also have no discernable mating preferences, potentially leading to F₂ individuals and/or backcrosses.

Significance statement

Several hybrid species are asexual, including a gynogenetic fish, *Poecilia formosa*, the Amazon molly. In this species the parental species are extant and can be crossed in the laboratory. Hybridization in nature suggests a breakdown of reproductive isolation. We explore this in our study on the behavioral level. We tested mating preferences of females of the two parental species, *P. mexicana* and *P. latipinna*, as well as F₁ females using binary choice tests. We found no statistically significant preferences, which may help explain the formation of the hybrids. Testing behavioral interactions of parental species is only possible for very few asexual hybrid species but comes with the caveat that since the formation of the original hybrids, the parental species have changed. Nonetheless, our study makes an interesting contribution to our understanding of hybridization in general.

Introduction

Hybridization between species in nature implies the absence or breakdown of reproductive isolating mechanisms (Rieseberg et al. 2000). These isolating mechanisms are thought to have evolved in response to costly hybridizations, especially for females. They are broadly classified into (i) mechanisms that prevent mating between two different species (prezygotic isolation) (Dagilis and Kirkpatrick 2016) and (ii) mechanisms leading to reduced fitness of hybrid offspring (postzygotic isolation) (Presgraves 2002). Hybridization is now recognized as an important factor in the evolution of species (Moran et al. 2021). Often this leads to introgression that is best detectable by genetic methods (Noonan et al. 2006; Cui et al. 2013; Powell et al. 2020). Full hybrid speciation, when hybrids are reproductively isolated from their parental species (Abbott et al. 2013; Runemark et al. 2019) appears to be much rarer, raising the question *why* hybrid species are so rare (or transient on an evolutionary time scale), when hybrid matings must be quite common. Alternatively, hybrid species may be quite common, but go undetected (Janko et al. 2018), partly because scientists are not systematically looking for them. In many cases, hybrid species show significant deviations from sexual reproduction, typically some form of clonal reproduction (Schlupp 2005; Avise 2008). These hybrid species with new reproductive modes are of particular interest to understand the evolutionary pathways that lead to the transition from sexuality to asexuality, as many asexual eukaryotes have been shown to be of hybrid origin (Suomalainen et al. 1987). In general, the resulting asexual modes of reproduction are genetically clonal, involve unreduced eggs, and sometimes require sperm to trigger embryogenesis (Avise 2008). The evolutionary pathway to emergence of these asexual species is still highly debated: though it is often thought that asexuality appears readily as a direct and immediate result of hybridization (Schlupp 2005), it has recently been proposed that hybrid contexts are fertile grounds for the evolution of asexuality through multiple small and incremental evolutionary steps (Fyon et al., submitted).

Several hypotheses to explain the rarity of asexual hybrids have been proposed. Moritz et al. (1989) suggested the ‘balance hypothesis’, postulating that asexuality should emerge within a certain range of genetic divergence between parentals. Within that range, accumulation of incompatibilities between parentals is enough to disrupt meiosis, but not viability and fertility in their hybrids (Moritz et al. 1989). Such a pattern was recently found in Whiptail lizards (Barley et

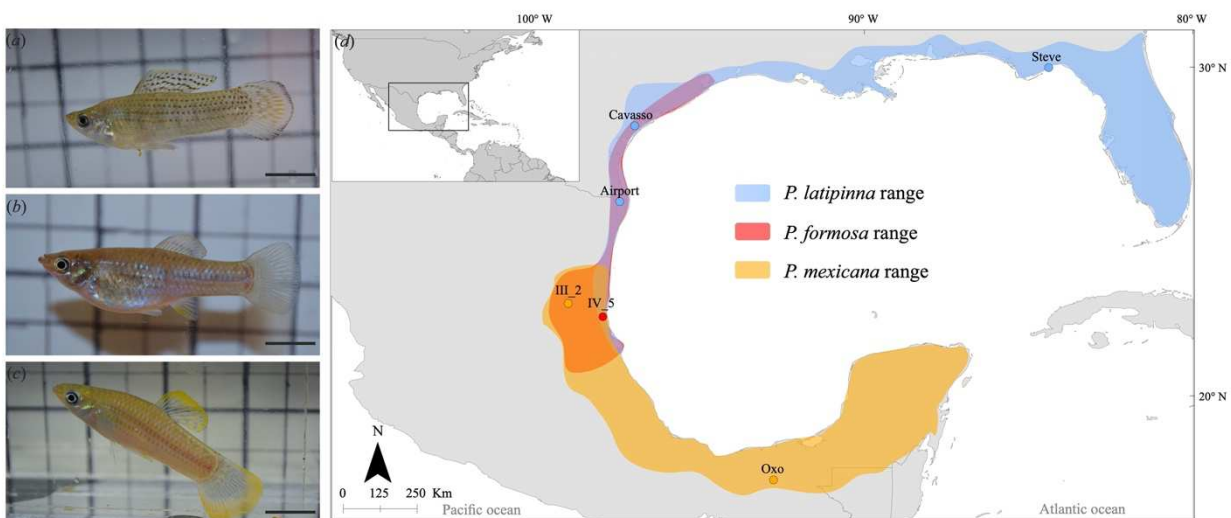
al. 2021; Barley et al. 2022). Using the Amazon molly (*Poecilia formosa*) as example, Stöck and colleagues (2010) proposed the ‘rare formation hypothesis’, suggesting that asexual hybrid are rare not due to inherent disadvantages of asexuality, but because a very specific genomic combination is needed to allow hybrid viability and clonal reproduction (Stöck et al. 2010; Warren et al. 2018). Freitas and colleagues (2022) are suggesting a slightly different view, where actual hybridizations are not rare, but most of them do not lead to the establishment of a stable hybrid lineage (Freitas et al. 2022). It should be noted that in species that are sperm-dependent (all asexual hybrid fishes and amphibians), the ecology of their establishment and maintenance is particularly complicated to explain in an evolutionary framework, given their need to coexist (but not outcompete) their sexual sperm donors.

Once established, asexual hybrids play important ecological roles, for example via resource competition with coexisting sexual species, or even in speciation of sexual species (Janko et al. submitted; Vrijenhoek 1994). Asexual Amazon mollies (*Poecilia formosa*), for example, show almost complete niche overlap with their sexual hosts, when it comes to feeding (Scharnweber et al. 2011a; Scharnweber et al. 2011b). In the present study we use the Amazon molly (*Poecilia formosa*), a unisexual hybrid species to investigate the role of mate choice in the formation of hybrid species. Hybridizations leading to unisexual species, often happened a long time ago (Avisé 2008) (in the case of the Amazon molly, at least 100.000 years (Schartl et al. 1995; Warren et al. 2018) and any role behavior may have played in the formation of hybrids is shrouded in history. In the case of the Amazon molly, however, the parental species are known and extant, allowing for a experiments that provide an approximation of the original hybridization. We are trying to shed light on a particularly poorly understood aspect of the origin of hybrid species and hybridization in general, namely the role of behavior. The study of female mate choice is particularly relevant for the origin of unisexual Amazon molly (*Poecilia formosa*), which originated via a hybridization event (Schlupp 2005; Lampert and Schartl 2008) between a *P. mexicana* “mother” and a *P. latipinna* “father” (Hubbs and Hubbs 1932). The hybrid origin has been confirmed several times through molecular work (Avisé et al. 1991; Schartl et al. 1995; Tiedemann et al. 2005; Warren et al. 2018). The Amazon molly is characterized by several uncommon traits. In addition to being a hybrid species, they have no meiosis (Monaco et al. 1984; Dedukh et al. 2022), are clonal, all-female, but require sperm to trigger embryogenesis. Uniquely, the two parental species that were involved in the single original hybridization that led to the

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Amazon molly have been identified and are extant, with the Atlantic molly *P. mexicana* (female parent) being distributed from South to Central Mexico, while the Sailfin molly *P. latipinna* (male parent) is present in coastal environments of the Gulf of Mexico and the Southeastern USA (Schlupp et al. 2002) (Figure 1).

Clearly, a hybrid species can only arise if heterospecific matings occur. This, however, is usually predicted to be selected against, as hybrid individuals are often less viable or fertile. There are examples, however, of adaptive heterospecific preference (Ryan and Wagner 1987; Pfennig 2007; Mendelson and Shaw 2012), sometimes governed by sensory bias (Ryan 1990; Basolo and Endler 1995). One reason why this problem may be difficult to address is that the parental species of a hybrid are often either unknown (Suomalainen et al. 1987), extinct (Freitas et al. 2019; Freitas et al. 2022), or otherwise unavailable. In a few cases, however, the parental species are known and extant. Only in such cases can we experimentally study the prerequisites for hybridization, namely mate choice for conspecific species. Of course, even when we have access to the parental species, one limitation is that the parental species have evolved between the time of hybridization that led to an asexual species and present time (Dries 2003; Makowicz and Travis 2020). Nonetheless, a study like the one presented here is uniquely important to shed light on the origin of unisexual hybrids. In the following, we ask whether females from two livebearing fishes from the family Poeciliidae (and their F_1 hybrids) with partially overlapping habitats show species preference as a prezygotic isolating mechanism.



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Figure 1: Males of *Poecilia latipinna* (a), *Poecilia mexicana* (c), and a female F₁ (b). The thick black bar in the lower right corner is a scale of one centimeter. The map depicts the current ranges of *Poecilia formosa* (orange and purple), *P. mexicana* (ochre) and *P. latipinna* (blue). In the orange area *P. formosa* coexists with *P. mexicana*, in the purple area is coexists with *P. latipinna*. The area of current sympatry of all three species is limited to the area around IV/5 (dark red). Circles indicate the populations used; note that we used populations from sympatry and allopatry. See text for details.

Furthermore, sympatry and syntopy are prerequisites for heterospecific matings to occur. Thus, for a meaningful test of mating preferences of the extant species, they should also overlap. Both of these conditions are met in the Amazon molly. A detailed analysis of mitochondrial haplotypes revealed that an area near Tampico, Mexico is the most likely place of origin of the Amazon molly (Stöck et al. 2010). This was independently supported by environmental niche modelling of the two parental species looking back in time (Costa and Schlupp 2020). Interestingly, this is also one of the few areas where both *P. mexicana* and *P. latipinna* currently coexist and are syntopic. This makes a particular population from a lagoon near Tampico, Mexico, a prime candidate for investigating the origin of Amazon mollies. It has already been used successfully in crossing *P. mexicana* and *P. latipinna* (Lampert et al. 2007), in addition to other populations (Hubbs 1934; Hubbs 1946a,b; Turner et al. 1980a; Dries 2003; Makowicz and Travis 2020).

Mating preferences between parental species can direct the evolutionary outcomes of hybridization. In the Amazon molly system, what is unknown so far is how likely the original hybridization would be with current individuals of *P. mexicana* and *P. latipinna*, especially from the syntopic populations from Tampico. The existence of the Amazon molly indicates that a unidirectional mating happened at least once, but how likely is it? Furthermore, why did the hybridization lead to Amazon mollies in only one direction? The attempts to cross the parental species are not informative here, because they were set up as forced crosses of one unmated (virgin) female and one or several males, leaving no opportunity for females to exercise mate choice. In general, mate choice in this system has been addressed (Schlupp 2005; Schlupp 2009; Schlupp 2018), including character displacement in males (Gabor and Ryan 2001; Gabor et al. 2005). Previous work on female preferences in mollies has converged on the idea of a pre-existing

bias governing (Ryan 1990; Endler and Basolo 1998) the mate preferences of female mollies, in particular females of species with short dorsal fins (Ptacek 1998), because males are more ornamented (Figure 1). For example, MacLaren and Rowland (2006) studied *P. mexicana* females from Rio Verde (likely from sympatry with *P. formosa*) and found that they prefer larger males (see also MacLaren et al. 2004), predicting heterospecific matings of such females with *P. latipinna* males. Note, however, that no Sailfin mollies are presently known from that area. Ptacek (1998) made similar observations for another shortfin molly species, *Poecilia orri*. If these preferences can be generalized, therefore, for the particular case of the origin of Amazon mollies this hypothesis predicts that a short fin molly female, such as *P. mexicana*, would be likely to mate with a long fin molly male, such as *P. latipinna*. In addition, this hypothesis predicts a directionality which is in agreement with the actual hybridization event and the directionality that led to the Amazon molly. However, a lack of preference and random mating by *P. mexicana* would also lead to hybridization, and the absence of asexual *de novo* hybrids in the wild with a long fin molly mother may be explained by unidirectional or sex-linked Muller-Dobzhansky incompatibilities (Schlupp 2005). Yet, both hypotheses, mating based on sensory bias, and random mating fail to account for the virtual absence of hybrids in nature. Asexual *de novo* hybrids are also unknown from laboratory crosses, instead all attempts to generate *de novo* asexual hybrids have resulted in sexual F₁'s (e.g. Hubbs 1934; Turner et al. 1980a; Dries 2003; Lampert et al. 2007; Makowicz and Travis 2020). Therefore, in our study, using a large dataset from multiple populations, we ask if females of *P. mexicana* and *P. latipinna* prefer conspecific males over heterospecific males. Furthermore, we study the mate preferences of a small number of F₁ females and ask if they have a preference for males of the parental species.

Because heterospecific matings are generally considered to be costly, we predict that females should prefer conspecific males. To address this, we used standard binary choice tests (Figure 2) and multiple populations of each species to account for within-species variability. Furthermore, we used unmated (virgin) and previously mated females in *P. mexicana* to elucidate the potential role of previous (or absence of) mating experience (Richardson and Zuk 2023). For this particular situation we predicted stronger preferences in females that had been together with conspecific males, because individual experience might reinforce their preference and the mated females had been together only with conspecifics.

In a subset of our experiments, we explored this paradigm deeper. We know that hybridizations leading to sexual F_1 happen – but apparently only in the laboratory (Hubbs 1934; Turner et al. 1980a; Dries 2003; Lampert et al. 2007; Makowicz and Travis 2020). On the other hand, a hybridization also happened in nature in the past, as evidenced by the existence of Amazon mollies. In other words, at some point in time likely F_1 females (possibly sexual, as in laboratory crosses) were faced with a choice between males of one or both of the two sexual species *P. latipinna* and *P. mexicana*, and F_1 males. For the first time to our knowledge, we were able to study female mate preferences in this situation. Male F_1 's were studied by Dries (2003). She tested if males of *P. latipinna* and *P. mexicana* would prefer F_1 hybrid females over gynogenetic Amazon mollies but found no differences. For our study, we would predict F_1 females to prefer F_1 males over sexual males if phenotypic similarity is a factor in mate preference.

Methods

Fishes and husbandry

Generally, fishes were kept in large stock tanks at the International Stock Center for Livebearing Fishes in a greenhouse located at the Aquatic Research Facility (ARF, 35.18325469955841, -97.4477283275434), which is maintained by the Department of Biology of the University of Oklahoma in Norman. Stock tanks are population specific and contain individuals of both sexes that are allowed to randomly outbreed. Tanks held either 500 or 1000 liters, contained natural plants and were connected to a flow-through system that is fed by well water. These large tanks contain a large amount of natural food which was supplemented with flake food (TetraMin) three times a week. Temperature was maintained at approximately 28 degrees but fluctuated somewhat with the temperature of the greenhouse. The light cycle was natural. Individuals from these stocks were moved into a dedicated fish-room in preparation for experiments. There they were sorted into sex specific 40-liter tanks and maintained until experimentation was completed. After that they were returned to their respective stock tanks for breeding. Consequently, non-virgin females originated from stock tanks where they were able to interact (and mate) with males from their own species and population. While it is highly likely that they did engage in prior matings, we cannot be sure of this. Also, their relatedness and familiarity with each other are not known.

Virgin (unmated) females were isolated in individual tanks at an age of about 30 days and subsequently raised individually in 9-liter tanks either in dedicated racks in the greenhouse or in a fish room at a 12:12h light-cycle and fed daily *ad libitum* with flake food, supplemented with frozen *Artemia* naupliae, *Daphnia* and mosquito larvae. Temperature in this room was maintained at 28 degrees.

Choice tests: Using virgin and non-virgin descendants of multiple populations of wild caught *P. mexicana* and multiple populations of non-virgin *P. latipinna*, including one wild caught population, individuals of each species were used in a typical choice experiment (Dougherty and Shuker 2014; Dougherty 2020) to test the species preference capabilities of the females of each species. In this, females were placed in a clear Plexiglas cylinder in the center of a 40-liter fish tank that was separated into thirds by lines drawn onto the tank (Figure 2).

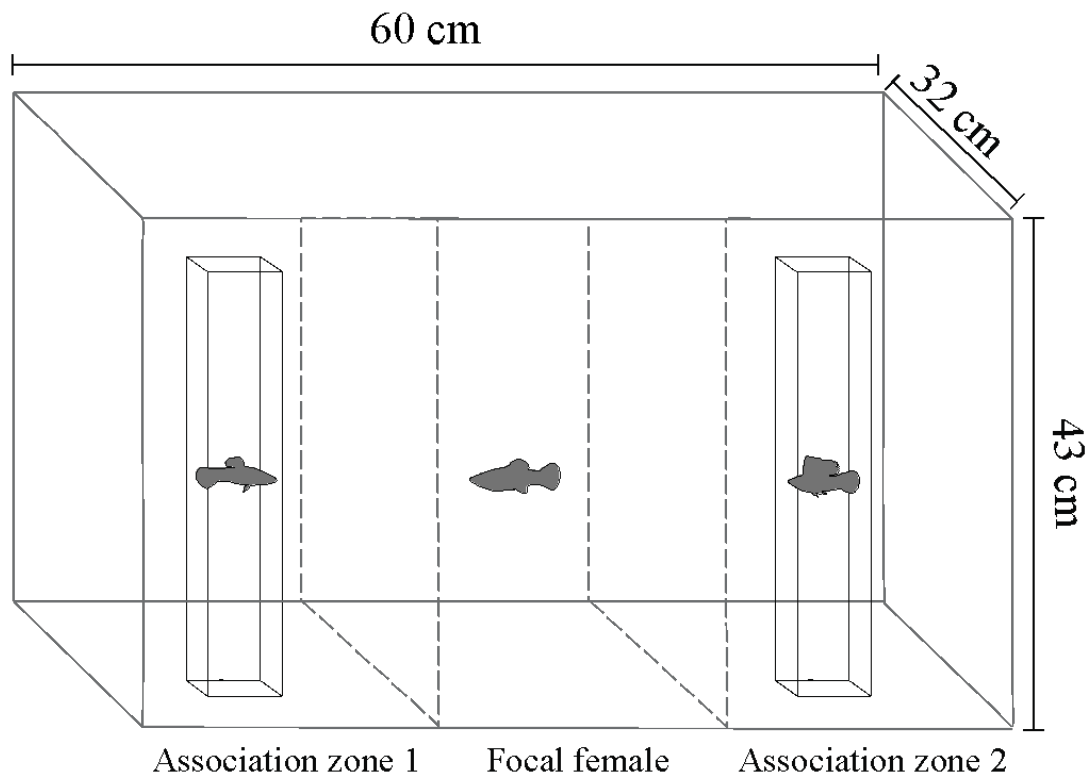


Figure 2: schematic of the choice tank with dimensions.

This configuration has been used successfully many times (for recent examples see Makowicz et al. 2016a; Makowicz et al. 2016b). Furthermore, association time is a widely used and accepted proxy for preference (reviewed in (Dougherty 2020)). For an experimental trial, a conspecific male and a heterospecific male were placed in their own cylinder, one in the left third of the tank and one in the right third. After 5 minutes of acclimating, the female was released from her cylinder and association time was recorded for a 5-minute interval. The association zones were the left or right third of the tank, respectively (Figure 2). The central area of the tank served as neutral zone, where time was not counted. The female was then gently recaptured, and the cylinders of the males were moved to the opposite side, to allow detecting potential side biases. We considered it a side bias when a chooser fish spent more than 80% of its time on one side of the test tank, without discernible reason. We recorded the number of side biases, tested if they were associated with a particular side of the tank and whether the females of different species differed in the frequency of occurrence. We removed data from side biased females from the analysis. Side biases are common in binary choice tests and *a priori* exclusion rules have been used before (Schlupp et al. 1994; Schlüter et al. 1998). Although our procedure controls statistically for side biases, we decided to exclude them because conceptually, individuals not responding to stimuli provide no meaningful preference information. However, the number of side biases can provide important information in itself, as high rates of side biases may be indicative of individuals responding to the provided stimuli in a meaningful way. We feel that even after removing data from side-biased individuals, our remaining sample size of 79 individuals is adequate. The female fish were then given another 5-minute acclimation period. After that, the female was again released, and association time was measured for another 5 minutes. Females were used only once in this experiment, but males had to be used more than once due to limited numbers of males of comparable size. Males were given a minimum of 24 hours of rest before reusing them. It was not possible to record data blindly because of logistical constraints; due to the complexity of the project, the observer had to be able to track individual females. We used 28 *P. latipinna* females and 51 *P. mexicana* females, 25 of which were unmated. Males were always matched for size. For population affiliation of these females, see Supplemental Table 1. Figure 1 provides a map depicting the geographic origin of the populations used as well as example photos of *P. mexicana*, *P. latipinna* males and a F₁ female.

Association time was measured using two stopwatches, one for each male. They were started every time the females mouth reached one of the thirds containing a male. After both halves of the trial were completed, the total time with each male was calculated and recorded. The position of the conspecific male was randomized before trials began using a random number generator that chose either a 1 or 2 with equal probability, meaning the conspecific was initially either on the left or right, respectively. The males in each trial were also measured for standard length (the tip of the snout to the caudal peduncle), so that they were within 1mm of each other to remove the potential influence of male size. Additionally, 30% of the water in the tank was replaced before each acclimation period.

Using F₁ individuals, the same choice experiment was conducted with a few alterations. Females in this experiment were used twice, once with another F₁ individual and a *P. latipinna* and once with another F₁ individual and a *P. mexicana*. The same ways to eliminate bias were utilized with an additional step of using the same random number generator to pick a number, 1 or 2, that determined whether the female would be tested with a male *P. mexicana* or the *P. latipinna* first, respectively. Trials using the same females were done with a minimum of 24 hours of rest for the fish and males were never paired with the same F₁ female to reduce individual bias as well.

All trials were video recorded using a Nikon digital camera; videos are available on request.

Statistical treatment

All statistical analyses were conducted in SPSS 28. All p-values are two-tailed. Specifically, we analyzed choice data with paired t-tests, and used a univariate GLM to explore the role of the species of the choosing individual in mate choice. The distribution of side biases was tested using a Chi² test.

Results

The sexual ancestors, Poecilia latipinna and Poecilia mexicana

a) Side biases

Overall, we tested 118 individual females of *P. mexicana* and *P. latipinna* from different populations (Supplementary Table 1). Several trials, 39 (33%), resulted in side-biases (Table 1).

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We tested whether the directionality or the species of chooser differed from random using a χ^2 test with Yates correction. Results indicated that the pattern did not significantly differ from random ($\chi^2=3.39$, $p=0.65$).

Table 1: number of side biases to the left or right per species

	Left	Right
<i>P. latipinna</i>	9	3
<i>P. mexicana</i>	10	17

b) Overall mate choice

After removing the 39 trials that showed a side bias from the dataset (assuming those females did not respond to the stimuli), we analyzed the remaining 79 trials for mate choice. Virgin females were used in 25 trials (54 were non-virgin), *P. mexicana* were used in 51 trials, *P. latipinna* in 28. We used six populations in our study, three each for the two species involved (Supplementary Table 1, Figure 1). Out of these three, one *P. mexicana* and one *P. latipinna* population each was from allopatry, the other two each from sympatry with *P. formosa*.

We tested for an overall preference for conspecific partners using a paired t-test comparing association times (time with heterospecific versus time with conspecific) (Figure 3). The mean time females spent with conspecific males was 210 (± 142 SD) seconds; they spent slightly less time (167 (± 115 SD) seconds) with heterospecific males. This is not significantly different ($t = 1.68$, $n = 79$, $p = 0.97$), in fact it is highly non-significant. We compared the sizes of the stimulus males, too, but there was no significant difference ($t = -0.4$, $n = 79$, $p = 0.67$). On average conspecific males were 26.92 (± 3.5) mm long, while heterospecific males were 26.96 (± 3.4) mm long. The females (or choosers) were on average 36.1 (± 7.8) mm long.

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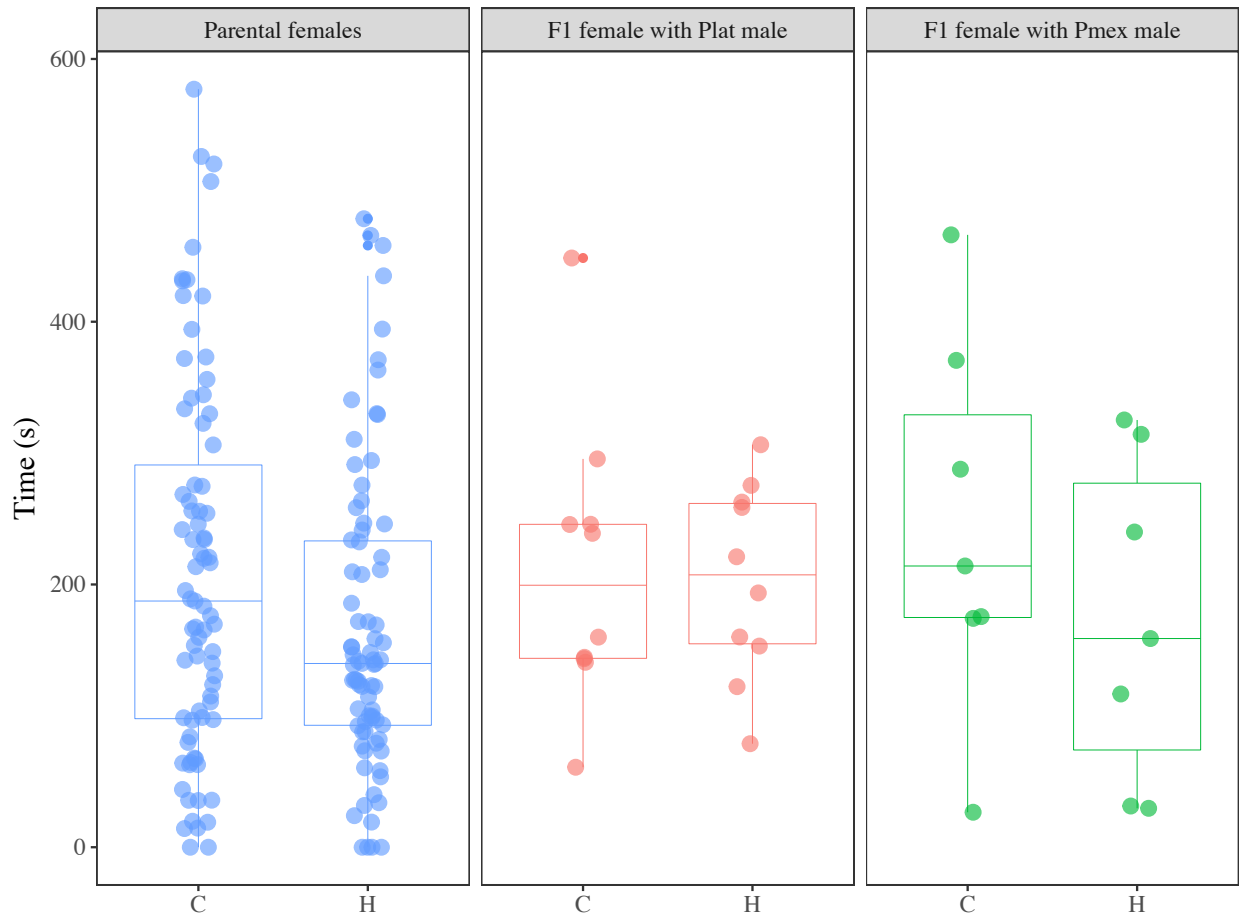


Figure 3: Boxplots of association times of the parental females (*Poecilia mexicana* and *Poecilia latipinna* combined), and the F₁ females with either a *P. mexicana* male or a *P. latipinna* male versus a F₁ hybrid male. C stands for conspecific, H for heterospecific.

Furthermore, we used a univariate GLM comparing the two species, *P. mexicana* and *P. latipinna*, with strength of preference (SOP), calculated as $((\text{time with the heterospecific male} + 1) / (\text{time with heterospecific male} + 1) + \text{time with conspecific male})$, as the response (dependent) variable. Size of the choosing female was used as a covariate. The order of presentation (whether the conspecific was first seen on the left or right side) was entered as a random factor. Because we had unequal sample sizes and the overall result was non-significant, we chose not to analyze effects at the population level (see supplemental material). As expected, because order of presentation had been randomized, order had no effect and was removed from the model (see supplemental material). The resulting GLM found no significant effect of species on preference, and a marginal effect of female (chooser) size (Species, DF 1, $F=1.416$, $p=.238$; size, DF 1, $F=3.893$, $P=0.052$).

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Based on the marginal effect of test fish size we ran an exploratory linear regression analysis but found no significant relationship between SOP and size ($t = -1.61$, $n = 79$, $p = 0.11$, $r = -0.18$, $r^2 = 0.03$).

c) *Poecilia mexicana* from Tampico only

For Amazon mollies, the natural hybrid of a *P. mexicana* mother and a *P. latipinna* father, we have a fair estimate of the locality in which the hybridization possibly occurred ca. 100.000 generations ago, southwest of what is now Tampico, México (Costa and Schlupp 2020) (Figure 1). We used females from a population that currently occurs in the general area which has been hypothesized to be historically syntopical and is known to be currently syntopical for the two parental species. We looked at this population separately using a paired t-test comparing the time in association with a *P. mexicana* and *P. latipinna* male. We were unable to detect a significant female preference of *P. mexicana* females for sympatric males ($t=1.137$, $p=0.285$, $n=10$); still, it is worth noting that on average females spent 60 seconds more with the conspecific *P. mexicana* male (*P. mexicana*: 185 seconds, *P. latipinna*: 124 seconds). We are aware, that we are only looking at a subset of the data and that this is nested within the general GLM already used for the overall comparison above.

d) Virgin females

Furthermore, some of the females we used were unmated (virgin), while the majority of females had previously been with males which is reflecting the most likely natural state. Because we only had access to unmated *P. mexicana*, mating status was not included into the overall model, but analyzed separately using a paired t-test comparing time with the conspecific (*P. mexicana*) versus the heterospecific (*P. latipinna*) male only for unmated females. There was no significant preference for conspecific males in unmated females ($t = 0.141$, $p = 0.889$, $n = 25$).

*F*₁ females

a) Side-biases

Other than the sexual ancestor females, we tested each *F*₁ female twice, once with a combination of an *F*₁ male and a *P. mexicana* male and with a combination of an *F*₁ male and a *P. latipinna* male. Out of the 24 trials we detected seven side biases (21.16%), two of which were for the right

side of the tank and five for the left side of the tank (Binomial test, $p=0.4531$). Most F_1 females showed a side bias only with one combination of males, but one female showed a side bias in both combinations. Removing the side biased trials thus created a statistical problem, because most cases affected only one of the two combinations, leaving us with an unbalanced dataset. To work around this problem, we analyzed the data for the two conditions separately.

b) Mate choice

When given a choice between an F_1 male and a *P. latipinna* male of the same size, F_1 females showed no discernable preference and spent roughly equal amounts of time with both males (paired t-test, $n=10$, $t=1.64$, $p=.871$) (Figure 3), namely 212.42 ± 108.42 SD seconds with the F_1 and 203.19 ± 74.00 SD seconds with the *P. latipinna* male. The choosing females were on average $29 \text{ mm} \pm 5.3$ long. F_1 males were on average $25.2 (\pm 3.22 \text{ mm})$ long and *P. latipinna* males $24.8 (\pm 2.39 \text{ mm})$ long.

A similar analysis was run with the choice data for F_1 females with F_1 males versus *P. mexicana* males (Figure 3). Also in this situation, the F_1 females showed no preference for either type of male (paired t- test, $n=7$, $t=0.72$, $p=0.5$) (Figure 2). F_1 females spent on average 245 ± 144.32 seconds with the F_1 male and 173.7 ± 123.55 seconds with the *P. mexicana* male. The choosing females were on average $19.86 \pm 6.6 \text{ mm}$ long, with the F_1 males being 26 ± 2.7 and the *P. mexicana* males being $26.14 \pm 2.97 \text{ mm}$ long.

Discussion

Overall mate choice and side biases: In our study we attempted to mimic the hypothetical situation that must have led to the hybrid origin of the unisexual Amazon molly, *P. formosa*. Based on conclusive genetic evidence we know that Amazon mollies originated a single time from a mating of a *P. mexicana* female and a *P. latipinna* male (Warren et al. 2018). Our findings show that such a mating would not have been hindered by strong mating preferences for conspecific males. This is in general agreement with previous studies that found that *P. mexicana* females prefer males with a *P. latipinna* phenotype (Ptacek 1998; MacLaren and Rowland 2006). Of course, this conclusion is only valid as much as mate preferences in *P. mexicana* have not changed much over the ca. 100.000 years of evolution that happened in the two sexual species since the formation of

the hybrid (Warren et al. 2018). Side biases occurred in ca. 33 % of the trials. This supports our conclusion that there are no detectable preferences in our study, as females with a side bias may be unable or unwilling to discern between the stimuli offered. Furthermore, virgin females did not differ in their preferences from non-virgin females, which supports findings from a meta-analysis that also found little effect of mating status across taxa (Richardson and Zuk 2023). However, in other studies, non-virgin and virgin females did differ in their preferences with non-virgin Sailfin mollies showing a preference for larger males, while virgin females showed no significant preference (Ptacek and Travis 1997).

We specifically tested individuals from a possible area of overlap, near Tampico, Mexico, but found no mating preference for conspecifics. We know from at least one laboratory study that *P. mexicana* and *P. latipinna* from that area can be successfully crossed to produce sexual F₁'s (Lampert et al. 2007) (Miron et al. unpublished data). More generally, in our study behavioral pre-zygotic isolation via female mate choice was detected. One conclusion from this is that heterospecific matings are not selected against. Based on other studies, heterospecific matings are even predicted (Ptacek 1998; MacLaren and Rowland 2006). However, while this may help explain the historical origin of the Amazon molly, it makes it difficult to understand the absence of natural F₁ hybrids in areas of overlap. We (I.S. with Manfred Scharl and Jakob Parzefall) and others have sampled in current areas of overlap (near Tampico, Mexico) and not detected any *de novo* F₁ hybrids (Stöck et al. 2010).

Given this and previous results, how can we explain the absence of natural F₁ hybrids in the wild? One possibility is that isolating mechanisms are in place, but female mate choice is not one of them. Other prezygotic isolating mechanisms may also play a role, but post-zygotic isolation may also be important. So far, we only know that F₁ individuals behaviorally differ from Amazon mollies (Dries 2003; Makowicz and Travis 2020), but a fitness comparison of F₁'s with any of the parental species has not yet been done. F₁'s do differ in some traits from parentals and have – for example – sex ratios that are female biased (Turner et al. 1980b; Lampert et al. 2007; Makowicz and Travis 2020); Berbel-Filho et al. unpublished data). Another possibility is that detecting hybrids in nature is generally difficult and unlikely (Contreras - Balderas 1990; Janko et al. 2018). Genetic work may be useful to overcome this limitation and has shown widespread

hybridization in livebearing fishes, for example in the genus *Xiphophorus* (Cui et al. 2013), and in many other taxa (Moran et al. 2021).

There are caveats, however, that need to be mentioned. The average size of the males and females used does not cover the natural size spectrum of the two species, especially the F_1 females were small. This was due to logistical constraints. Although, this may have led to weaker female preferences, choice between two small males is clearly a natural situation. Furthermore, preferences of males can be influenced by the timing of the experiments relative to the hormonal cycle of the choosing females (Parzefall 1973). Both male and female choice is more acute when females are in the short phase of their sexual cycle, when their eggs can be fertilized. However, in many studies females – often wild caught - with unknown mating status were used, and preferences are commonly detected (e.g. Ptacek and Travis 1997). Our findings would be less surprising if mollies generally showed little or no preference for other mating related traits, but this is not the case. In both species, *P. mexicana* and *P. latipinna* – female (Schlupp 2009) and male preferences (Schlupp 2018; Schlupp 2021) for several traits have been documented. Males show preference for sexual females over Amazon mollies (Schlupp et al. 1994), *P. latipinna* females show a preference for larger mates (Ptacek and Travis 1997; Witte and Ryan 1998; Gabor and Page 2003), and more symmetrical males (Schlüter et al. 1998). In *P. mexicana* essentially the same pattern is found (Jordan et al., 2006; Plath et al., 2004; Sommer-Trembo et al., 2020; Zimmer et al., 2018). Together these studies mirror more general findings in the family of Livebearing fishes, where on average females prefer more ornamented, larger males (Rios-Cardenas and Morris 2011).

F₁ females: The absence of a detectable preference in F_1 hybrids is not surprising given the lack of preference in parental species. We had predicted that the preference of the F_1 's might have an inherited preference similar to that of the maternal ancestor, and this may indeed be the case. Our data does predict, however, that after the occurrence of sexual F_1 's, indiscriminate mating may have led to a hybrid swarm. This was suggested by (Alberici da Barbiano et al. 2013) as an early stage of the evolution of the Amazon molly. This pathway, however, is not compatible with genomic data showing a single hybrid origin of the Amazon molly (Warren et al. 2018). Nonetheless, this single origin does not rule out the idea of complex interactions between parental species, F_1 's and beyond.

In the bigger picture it seems important to evaluate the role of mate preferences in hybridization more generally. Clearly in the case of the Amazon molly, some breakdown of isolation must have happened. We know this through the existence of Amazon mollies, but other such breakdowns may not leave behind such a large footprint (Janko et al. 2018). Indeed, while reports of actual hybrids between mollies in nature are rare, there is mounting evidence for a bigger role of hybridization than previously assumed (Schwenk et al. 2008; Moran et al. 2021). Furthermore, even in the presence of clear preferences, heterospecific matings may occur. This was argued in a study by McCoy et al. (2011), that investigated preferences for a novel trait, a mustache (Schlupp et al. 2010), that is only present in one of the species studied. Only that species, *Poecilia sphenops*, showed a significant preference for the mustache, but females from several other species showed some preference for the novel trait (McCoy et al. 2011).

Traditionally, hybridization is considered to be costly for females, and generally selected against, especially if the rate of overlap of the species is very limited and choice is rarely under selection. However, mounting evidence shows that hybridization is widespread, and plays an important role in species evolution (Mallet 2007). One excellent example is the origin of a new hybrid species in Galapagos Finches (Grant and Grant 1992). In this case, the hybridization was directly observed, a situation that is highly unlikely, yet illustrates the crucial role of behavior in the origin of hybrids. Genetic studies are capable of providing evidence for such hybridizations (Lamichhaney et al. 2018) but cannot elucidate the behavioral processes that allow for hybridizations to happen. Our study may provide an example for the absence or a breakdown of prezygotic isolating mechanisms, allowing a hybrid species to appear and flourish.

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Declarations

Ethics approval

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Animal experimentation was granted by the University of Oklahoma IACUC under protocol numbers R19-011 and R21-008. Permission to collect fishes in Texas was granted by Texas Parks and Wildlife # SPR-0320-034.

Data availability

The data analyzed for this study is available in the supplementary material. Video recordings of the trials are available on request.

Conflict of interest

The authors declare no conflict of interest.

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